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Hormonal Control During Exercise

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In Closing



On May 22, 2010, a 13-year-old American boy became the youngest climber to reach the top of Mount Everest, a grueling trek to an altitude 29,035 ft (8,850 m) above sea level. The climb was extremely controversial because of the boy's age. In fact, because the Nepalese government would not give the family permission to climb Everest from Nepal, the climbing team ascended from the more difficult Chinese side where there was no age restriction. To prepare for the climb, the boy and his father (and climbing partner) slept for months in a hypoxic tent to prepare their bodies for ascent to high altitude. One goal of high-altitude acclimation is to increase the concentration of oxygen-carrying red blood cells in the blood. Two important hormones facilitated this goal. An increase in the hormone erythropoietin signaled the bone marrow to produce more red blood cells, and a decrease in vasopressin (also called antidiuretic hormone) caused the kidneys to produce excess urine to better concentrate the red blood cells. Because of these adaptations, the climbers were able to summit Mount Everest with less time spent in the various base camps along the way.

During exercise and exposure to extreme environments, the body must make a multitude of physiological adjustments. Energy production must increase, and metabolic by-products must be cleared. Cardiovascular and respiratory function must be constantly adjusted to match the demands placed upon these and other body systems, such as those regulating temperature. While the body's internal environment is in a constant state of flux even at rest, during exercise these well-orchestrated changes must occur rapidly and in a well-coordinated manner.

While much of the physiological regulation and integration required during exercise is accomplished by the nervous system (discussed in [chapter 3](#)), another physiological system—the endocrine system—affects virtually every cell, tissue, and organ in the body. It constantly monitors the body's internal environment, noting all changes that occur and rapidly releasing hormones to ensure that homeostasis is not dramatically disrupted. In this chapter, we focus on the importance of hormones in maintaining homeostasis and aiding all the internal processes that support physical activity. Because we cannot cover all aspects of endocrine control during exercise, the focus is on hormonal control of metabolism and body fluid balance

during exercise. Because diet plays an important role in exercise metabolism, hormonal regulation of food intake is also covered. Additional hormones—including those that regulate growth and development, muscle mass, and reproductive function—are covered in other chapters of this book.

The Endocrine System

As the body transitions from a resting to an active state, the rate of metabolism must increase to provide necessary energy. This requires the coordinated integration and communication of many physiological and biochemical systems. Although the nervous system is responsible for much of this communication, fine-tuning the physiological responses to any disturbance in homeostasis is primarily the responsibility of the endocrine system. The endocrine and nervous systems, often collectively called the neuroendocrine system, work in concert to control all of the physiological processes that support exercise. The nervous system functions quickly, having short-lived, localized effects, whereas the endocrine system responds more slowly but has longer-lasting effects.

The endocrine system is defined as all tissues or glands that secrete **hormones**. The major endocrine glands and tissues are illustrated in [figure 4.1](#). Endocrine glands typically secrete their hormones directly into the blood where they act as chemical signals throughout the body. When secreted by the specialized endocrine cells, hormones are transported via the blood to specific **target cells**—cells that possess specific hormone receptors. On reaching their destinations, hormones can control the activity of the target tissue. Historically, hormones were defined as chemicals made by a gland that traveled to a remote tissue in the body to exert their action. Now hormones are more broadly defined as any chemical that controls and regulates the activity of certain cells or organs. Some hormones affect many body tissues, including the brain, whereas others target specific cells within a tissue.

Hormones are involved in most physiological processes, so their actions are relevant to many aspects of exercise and physical activity. Because hormones play key roles in almost every system of the body, total coverage of that topic is well beyond the scope of this book. In

the following sections, the chemical nature of hormones and the general mechanisms through which they act are discussed. An overview of the major endocrine glands and their hormones is presented for completeness. With respect to exercise, the focus is on two major aspects of hormonal control, the control of exercise metabolism and the regulation of body fluids and electrolytes during exercise. Finally, new information about hormonal regulation of food intake is presented, since caloric intake and specific nutrients consumed have a profound influence on exercise metabolism.

Chemical Classification of Hormones

Hormones are traditionally categorized as steroid hormones and nonsteroid hormones. **Steroid hormones** have a chemical structure similar to cholesterol, since most are derived from cholesterol. For this reason, they are soluble in lipids so they diffuse rather easily through cell membranes. This group includes the reproductive hormones testosterone (secreted by the testes) and estrogen and progesterone (secreted by the ovaries and placenta), as well as cortisol and aldosterone (secreted by the adrenal cortex).

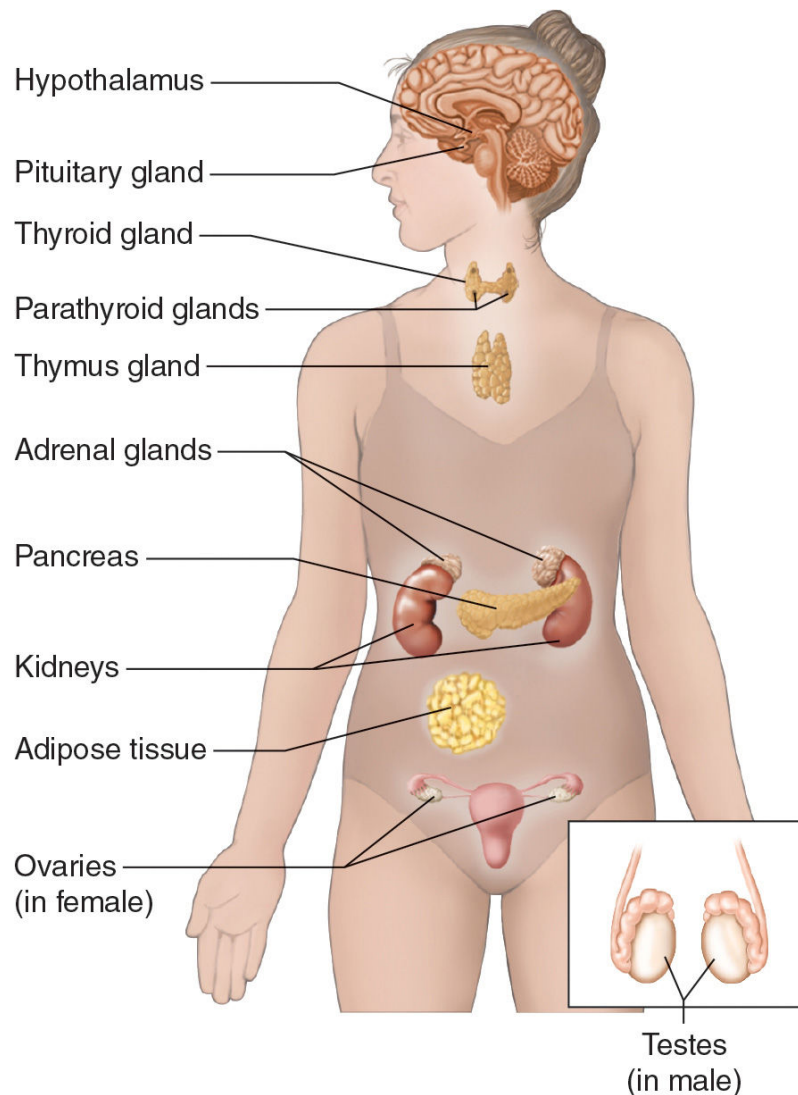


FIGURE 4.1 Location of the major endocrine organs of the body.

Nonsteroid hormones are not lipid soluble, so they cannot easily cross cell membranes. The nonsteroid hormone group can be subdivided into two groups: protein or peptide hormones and amino acid–derived hormones. The two hormones produced by the thyroid gland (thyroxine and triiodothyronine) and the two from the adrenal medulla (epinephrine and norepinephrine) are amino acid–derived hormones. All other nonsteroid hormones are protein or peptide hormones. The chemical structure of a hormone determines its mechanism of action on target cells and tissues.

Hormone Secretion and Plasma Concentration

Control of hormone secretion must be rapid in order to meet the demands of changing bodily functions. Hormones are not secreted constantly or uniformly, but often in a pulsatile manner, that is, in irregularly timed brief bursts. Therefore, plasma concentrations of specific hormones fluctuate over short periods of an hour or less. But plasma concentrations of many hormones also fluctuate over longer periods of time, showing daily or even monthly cycles (such as monthly menstrual cycles). How do endocrine glands know when to release their hormones and how much to release?

Negative feedback is the primary mechanism through which the endocrine system maintains homeostasis. Secretion of a hormone causes some change in the body, and this change in turn inhibits further hormone secretion. Consider how a home thermostat works. When the room temperature decreases below some preset level, the thermostat signals the furnace to produce heat. When the room temperature increases to the preset level, the thermostat's signal ends, and the furnace stops producing heat. In the body, secretion of a specific hormone is similarly turned on or off (or up or down) by specific physiological changes.

Using the example of plasma glucose concentrations and the hormone insulin, when the plasma glucose concentration is high, the pancreas releases insulin. Insulin increases cellular uptake of glucose, lowering plasma glucose concentration. When plasma glucose concentration returns to normal, insulin release is inhibited until the plasma glucose level increases again. Because the endocrine system works in concert with the nervous system, the central nervous system is also involved in maintenance of appropriate hormonal balance.

The plasma concentration of a specific hormone is not always the best indicator of that hormone's activity because hormones must bind to specific cellular receptors to exert an effect. Accordingly, the number of receptors on target cells can be altered to increase or decrease that cell's sensitivity to the hormone. With fewer receptors, fewer hormone molecules can bind, and the cell becomes less sensitive to the given hormone. This is referred to as **downregulation**, or desensitization. In people with **insulin resistance**, for example, the number of insulin receptors on their

cells appears to be reduced. Their bodies respond by increasing insulin secretion from the pancreas, so their plasma insulin concentrations increase. To obtain the same degree of plasma glucose control as normal, healthy people, these individuals must release much more insulin.

In a few instances, a cell may respond to the prolonged presence of large amounts of a hormone by increasing its number of available receptors. When this happens, the cell becomes more sensitive to that hormone because more can be bound at one time. This is referred to as **upregulation**. For example, individuals with a high **insulin sensitivity**, the opposite of insulin resistance, need relatively normal or low levels of insulin to process a given concentration of blood glucose.

Hormone Actions

Because hormones travel in the blood, they contact virtually all body tissues. How, then, do they limit their effects to specific targets? This ability is attributable to the specific hormone receptors on target tissues that can bind only specific hormones. Each cell typically has from 2,000 to 10,000 receptors. The combination of a hormone and its bound receptor is referred to as a hormone–receptor complex.

Recall that steroid hormones are lipid soluble and can therefore pass through cell membranes whereas nonsteroid hormones cannot. Receptors for nonsteroid hormones are located on the cell membrane, while those for steroid hormones are found either in the cytoplasm or in the nucleus of the cell. Each hormone is usually highly specific for a single type of receptor and binds only with its specific receptors, thus affecting only tissues that contain those specific receptors. Once hormones are bound to a receptor, numerous mechanisms allow them to control the actions of those cells.

Steroid Hormones

The general mechanism of action of steroid hormones is illustrated in [figure 4.2](#). Once through the cell membrane and inside the cell, a steroid hormone binds to its specific receptors. The hormone–receptor complex then enters the nucleus, binds to part of the cell's DNA (deoxyribonucleic acid), and activates certain genes. This

process is referred to as **direct gene activation**. In response to this activation, mRNA (messenger ribonucleic acid) is synthesized within the nucleus. The mRNA then enters the cytoplasm and promotes protein synthesis. These proteins may be

- enzymes that can have numerous effects on cellular processes,
- structural proteins for tissue growth and repair, or
- regulatory proteins that can alter enzyme function.

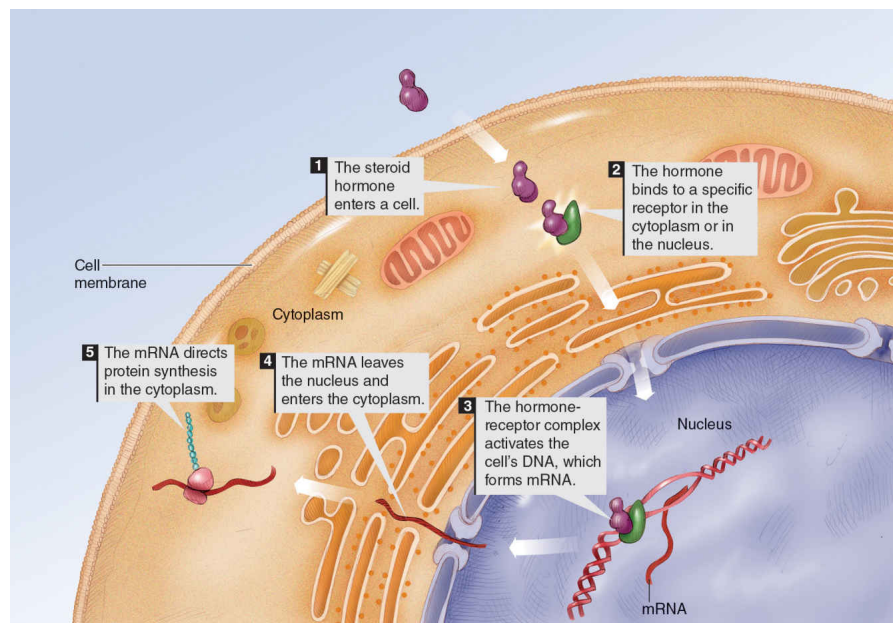


FIGURE 4.2 The general mechanism of action of a typical steroid hormone, leading to direct gene activation and protein synthesis.



Nonsteroid Hormones

Because nonsteroid hormones cannot cross the cell membrane, they bind with specific receptors on the cell membrane. A nonsteroid hormone molecule binds to its membrane receptor and triggers a series of reactions that lead to the formation of an intracellular **second messenger**. In addition to relaying signals, second messengers can also help intensify the strength of the signal. While there are many second messenger molecules, one important second messenger that mediates multiple hormone–receptor responses is **cyclic adenosine monophosphate (cAMP)**, or cyclic AMP); its mechanism of action is depicted in [figure 4.3](#). In this case, attachment of the hormone to the appropriate membrane receptor activates an enzyme, adenylate cyclase, situated within the cell membrane. This enzyme regulates the formation of cAMP from cellular adenosine triphosphate (ATP). Cyclic AMP then controls specific physiological responses that can include

- activation of cellular enzymes,
- change in membrane permeability,
- promotion of protein synthesis,
- change in cellular metabolism, or
- stimulation of cellular secretions.

Some of the hormones that employ cAMP as a second messenger are epinephrine, glucagon, and luteinizing hormone. In addition to cAMP, other important second messengers include cyclic guanosine monophosphate (cGMP), inositol trisphosphate (IP3), diacylglycerol (DAG), and calcium ions (Ca^{2+}).

Although by strict definition not hormones, **prostaglandins** are often considered to be a third class of hormones. These substances are derived from a fatty acid, arachidonic acid, and they are associated with the plasma membranes of almost all body cells. Prostaglandins typically act as *local* hormones or **autocrines**, exerting their effects in the immediate area where they are produced. But some also survive long enough to circulate through the blood to affect distant tissues. Prostaglandin release can be triggered by many stimuli, such as other hormones or a local injury. Their functions are quite numerous because there are several different types of prostaglandins. They often mediate the effects of other hormones.

They are also known to act directly on blood vessels, increasing vascular permeability (which promotes swelling) and vasodilation. In this capacity, they are important mediators of the inflammatory response. They also sensitize the nerve endings of pain fibers; thus, they mediate both inflammation and pain.

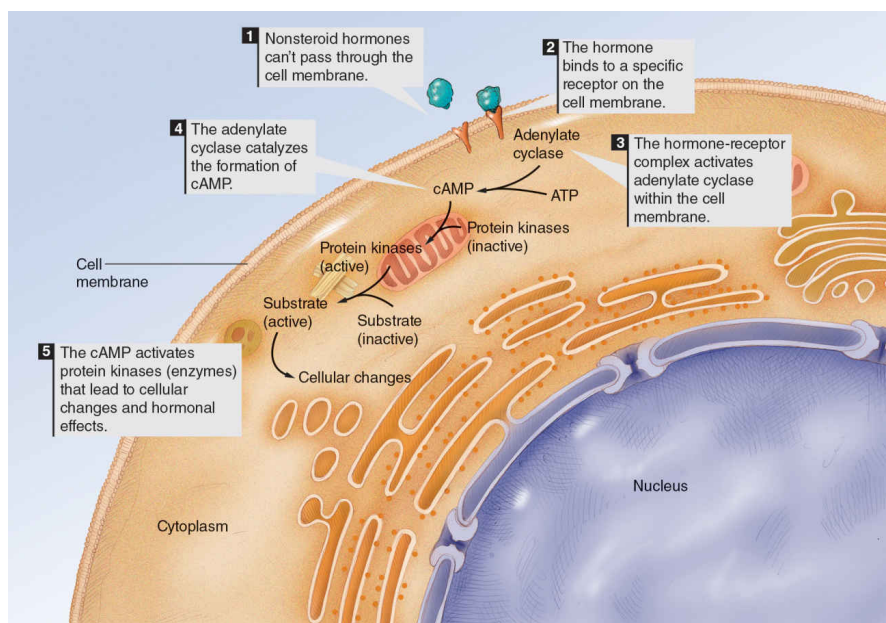


FIGURE 4.3 The mechanism of action of a nonsteroid hormone, in this case activating a second messenger (cyclic adenosine monophosphate) within the cell to activate cellular functions.



Endocrine Glands and Their Hormones: An Overview

The major endocrine glands and their respective hormones are listed in [table 4.1](#). This table also lists each hormone's primary target and actions. Because the endocrine system is extremely complex, the

presentation here has been greatly simplified to focus on those endocrine glands and hormones of greatest importance to exercise and physical activity.

Because hormones play such an important role in regulation of many physiological variables during exercise, it is not surprising that hormone release changes during acute bouts of activity. The hormonal responses to an acute bout of exercise and to exercise training are summarized in [table 4.2](#). This table is limited to those hormones that play major roles in sport and physical activity. Further details of these exercise-induced hormonal responses are provided in the following discussion of specific endocrine glands and their hormones.

TABLE 4.1 The Major Endocrine Glands and Their Hormones, Target Organs, Controlling Factors, and Functions

Endocrine gland	Hormone	Target organ	Controlling factor	Major functions
Anterior pituitary	Growth hormone (GH)	All cells in the body	Hypothalamic GH-releasing hormone; GH-inhibiting hormone (somatostatin)	Promotes development and enlargement of all body tissues until maturation; increases rate of protein synthesis; increases mobilization of fats and use of fat as an energy source; decreases rate of carbohydrate use
	Thyrotropin (TSH)	Thyroid gland	Hypothalamic TSH-releasing hormone	Controls the amount of thyroxine and triiodothyronine produced and released by the thyroid gland
	Adrenocorticotropin (ACTH)	Adrenal cortex	Hypothalamic ACTH-releasing hormone	Controls the secretion of hormones from the adrenal cortex
	Prolactin	Breasts	Prolactin-releasing and -inhibiting hormones	Stimulates milk production by the breasts
	Follicle-stimulating hormone (FSH)	Ovaries and testes	Hypothalamic FSH-releasing hormone	Initiates growth of follicles in the ovaries and promotes secretion of estrogen from the ovaries; promotes development of the sperm in the testes
	Luteinizing hormone (LH)	Ovaries and testes	Hypothalamic FSH-releasing hormone	Promotes secretion of estrogen and progesterone and causes the follicle to rupture, releasing the ovum; causes testes to secrete testosterone
Posterior pituitary	Antidiuretic hormone (ADH or vasopressin)	Kidneys	Hypothalamic secretory neurons	Assists in controlling water excretion by the kidneys; elevates blood pressure by constricting blood vessels
	Oxytocin	Uterus and breasts	Hypothalamic secretory neurons	Controls contraction of uterus; milk secretion
Thyroid	Thyroxine (T_4) and triiodothyronine (T_3)	All cells in the body	TSH, T_3 , and T_4 concentrations	Increase the rate of cellular metabolism; increase rate and contractility of the heart
	Calcitonin	Bones	Plasma calcium concentrations	Controls calcium ion concentration in the blood

continued

Endocrine gland	Hormone	Target organ	Controlling factor	Major functions
Parathyroid	Parathyroid hormone (PTH, or parathormone)	Bones, intestines, and kidneys	Plasma calcium concentrations	Controls calcium ion concentration in the extracellular fluid through its influence on bones, intestines, and kidneys
Adrenal medulla	Epinephrine	Most cells in the body	Baroreceptors, glucose receptors, and brain and spinal centers	Stimulates breakdown of glycogen in liver and muscle and lipolysis in adipose tissue and muscle; increases skeletal muscle blood flow; increases heart rate and contractility; increases oxygen consumption
	Norepinephrine	Most cells in the body	Baroreceptors, glucose receptors, and brain and spinal centers	Stimulates lipolysis in adipose tissue and in muscle to a lesser extent; constricts arterioles and venules, thereby elevating blood pressure
Adrenal cortex	Mineralocorticoids (aldosterone)	Kidneys	Angiotensin and plasma potassium concentrations; renin	Increase sodium retention and potassium excretion through the kidneys
	Glucocorticoids (cortisol)	Most cells in the body	ACTH	Control metabolism of carbohydrates, fats, and proteins; exert an anti-inflammatory action
	Androgens and estrogens	Ovaries, breasts, and testes	ACTH	Assist in the development of female and male sex characteristics
Pancreas	Insulin	All cells in the body	Plasma glucose and amino acid concentrations	Controls blood glucose levels by lowering glucose levels; increases use of glucose and synthesis of fat
	Glucagon	All cells in the body	Plasma glucose and amino acid concentrations	Increases blood glucose; stimulates the breakdown of protein and fat
	Somatostatin	Islets of Langerhans and intestines	Plasma glucose, insulin, and glucagon concentrations	Depresses the secretion of both insulin and glucagon
Kidney	Renin	Adrenal cortex	Plasma sodium concentrations	Assists in blood pressure control
	Erythropoietin (EPO)	Bone marrow	Low tissue oxygen concentrations	Stimulates erythrocyte production
Testes	Testosterone	Sex organs and muscle	FSH and LH	Promotes development of male sex characteristics, including growth of testes, scrotum, and penis, facial hair, and change in voice; also promotes muscle growth
Ovaries	Estrogens and progesterone	Sex organs and adipose tissue	FSH and LH	Promote development of female sex organs and characteristics; increase storage of fat; assist in regulating the menstrual cycle

TABLE 4.2 Hormone Responses to Acute Exercise and Change in Response With Exercise Training

Endocrine gland	Hormone	Response to acute exercise (untrained)	Effect of exercise training
Anterior pituitary	Growth hormone (GH)	Increases with increasing rates of work	Attenuated response at same rate of work
	Thyrotropin (TSH)	Increases with increasing rates of work	No known effect
	Adrenocorticotropin (ACTH)	Increases with increasing rates of work and duration	Attenuated response at same rate of work
	Prolactin	Increases with exercise	No known effect
	Follicle-stimulating hormone (FSH)	Small or no change	No known effect
Posterior pituitary	Luteinizing hormone (LH)	Small or no change	No known effect
	Antidiuretic hormone (ADH or vasopressin)	Increases with increasing rates of work	Attenuated response at same rate of work
Thyroid	Oxytocin	Unknown	Unknown
	Thyroxine (T ₄) and triiodothyronine (T ₃)	Free T ₃ and T ₄ increase with increasing rates of work	Increased turnover of T ₃ and T ₄ at same rate of work
Parathyroid	Calcitonin	Unknown	Unknown
	Parathyroid hormone (PTH or parathormone)	Increases with prolonged exercise	Unknown
Adrenal medulla	Epinephrine	Increases with increasing rates of work, starting at about 75% of $\dot{V}O_{2max}$	Attenuated response at same rate of work
	Norepinephrine	Increases with increasing rates of work, starting at about 50% of $\dot{V}O_{2max}$	Attenuated response at same rate of work
Adrenal cortex	Aldosterone	Increases with increasing rates of work	Unchanged
	Cortisol	Increases only at high rates of work	Slightly higher values
Pancreas	Insulin	Decreases with increasing rates of work	Attenuated response at same rate of work
	Glucagon	Increases with increasing rates of work	Attenuated response at same rate of work
Kidney	Renin	Increases with increasing rates of work	Unchanged
Testes	Erythropoietin (EPO)	Unknown	Unchanged
	Testosterone	Small increases with exercise	Resting levels decreased in male runners
Ovaries	Estrogens and progesterone	Small increases with exercise	Resting levels might be decreased in highly trained women

As mentioned earlier, a comprehensive description of neuroendocrine control is well beyond the scope of this textbook. Two important exercise-related functions of the endocrine glands and their hormones are the regulation of metabolism during exercise and the regulation of body fluids and electrolytes. The endocrine system also plays an important role in regulating appetite and food intake. The sections that follow detail these three important functions. Each section provides a description of the primary endocrine glands involved, the hormones produced, and how those hormones serve the given regulatory role.

VIDEO 4.1 Presents Katarina Borer on the contributing role of sex hormones to ACL tears in women.



In Review

- The nervous system functions quickly, having short-lived, localized effects, whereas the endocrine system typically responds more slowly but has longer-lasting effects.
- Hormones are classified chemically as either steroid or nonsteroid. Steroid hormones are lipid soluble, and most are formed from cholesterol. Nonsteroid hormones are formed from proteins, peptides, or amino acids.
- Hormones influence specific target tissues or cells through a unique interaction between the hormone and the specific receptors for that hormone on the cell membrane (nonsteroid hormones) or within the cytoplasm or nucleus of the cell (steroid hormones).
- Hormones generally are secreted nonuniformly, often in brief pulsatile bursts, into the blood and then circulate to target cells.
- A negative feedback system regulates secretion of most hormones.
- The number of receptors for a specific hormone can be altered to meet the body's demands. *Upregulation* refers to an increase in available receptors, and *downregulation* refers to a decrease. These two processes change a cell's sensitivity to a given hormone.
- Steroid hormones pass through cell membranes and bind to receptors in the cytoplasm or nucleus of the cell. At the nucleus, they use a mechanism called direct gene activation to cause protein synthesis.
- Nonsteroid hormones cannot easily enter cells, so they bind to receptors on the cell membrane. This activates a second messenger within the cell, often cAMP, which in turn can trigger numerous cellular processes.
- Prostaglandins are not hormones by strict definition but act as local hormones, exerting their effect in the immediate area where they are produced.

Hormonal Regulation of Metabolism During Exercise

As noted in [chapter 2](#), carbohydrate and fat metabolism are responsible for maintaining muscle ATP during prolonged exercise. Various hormones work to ensure adequate glucose and free fatty acid (FFA) availability for muscle energy metabolism. In the next sections, we examine (1) the major endocrine glands and hormones responsible for metabolic regulation and (2) how the metabolism of glucose and fat is regulated by these hormones during exercise.

Endocrine Glands Involved in Metabolic Regulation

While many complex systems interact to regulate metabolism at rest and during exercise, the major endocrine glands responsible are the anterior pituitary gland, the thyroid gland, the adrenal glands, and the pancreas.

Anterior Pituitary

The pituitary gland is a marble-sized gland attached to the hypothalamus at the base of the brain. It has three lobes: anterior, intermediate, and posterior. The intermediate lobe is very small and is thought to play little or no role in humans, but both the anterior and posterior lobes serve major endocrine functions. Hormonal release from the anterior pituitary is controlled by hormones secreted by the hypothalamus, while the posterior pituitary releases hormones in response to direct nerve signals from the hypothalamus. Therefore, the pituitary gland can be thought of as the relay between CNS control centers and peripheral endocrine glands. The posterior pituitary is discussed later in the chapter.

The anterior pituitary, also called the adenohypophysis, secretes six hormones in response to **releasing factors** or **inhibiting factors** (which are also categorized as hormones) secreted by the hypothalamus. Hormonal communication between the hypothalamus and the anterior lobe of the pituitary occurs through a specialized circulatory system. The major functions of each of the anterior pituitary hormones, along with their releasing and inhibiting factors, are listed in [table 4.1](#). Exercise is a strong stimulus to the

hypothalamus because exercise increases the release of most anterior pituitary hormones (see [table 4.2](#)).

Of the six anterior pituitary hormones, four are tropic hormones, meaning they affect the functioning of other endocrine glands. The exceptions are growth hormone and prolactin. **Growth hormone (GH)** is a potent anabolic agent (a substance that builds up organs and tissues, producing growth and cell differentiation and an increase in size of tissues). It promotes muscle growth and hypertrophy by facilitating amino acid transport into the cells. In addition, GH directly stimulates fat metabolism (lipolysis) by increasing the synthesis of lipolytic enzymes. Growth hormone concentrations are elevated during both aerobic and resistance exercise in proportion to the exercise intensity and typically remain elevated for some time after exercise.

RESEARCH PERSPECTIVE 4.1

Does Having More Testosterone Give You a Competitive Advantage?

Androgens (testosterone and its chemical derivatives) stimulate the development and maintenance of primary and secondary male sex characteristics. Although androgens are typically described as male sex hormones, they are found naturally in both men and women and can improve sport performance in both male and female athletes, particularly in strength-dependent events. Because of their ergogenic effects (enhanced physical performance, stamina, and recovery), androgens have been widely abused by athletes despite advances in tests to detect their abuse. In fact, androgens are the most common ergogenic aid used by female athletes. However, some women have naturally higher circulating androgens, and a great deal of controversy has surrounded the debate about whether these women should be allowed to compete with this natural ergogenic advantage. Because of this controversy, regulatory committees are keenly interested in scientific evidence that may link circulating natural androgens and athletic performance.

A recent study of 2,127 elite track and field athletes competing in the 2011 and 2013 International Association of Athletics Federations World Championships provided more scientific data on this controversial topic.¹ Researchers measured blood androgens, particularly testosterone concentrations, in male and female athletes and compared these concentrations to each athlete's best performances at the World Championships. Male sprinters showed higher testosterone concentrations, and men involved in throwing events had lower testosterone concentrations

than male athletes in other events. The type of event had no association with testosterone concentration in women. However, women (but not men) with the highest testosterone concentrations performed better in the 400 m, 400 m hurdles, 800 m, hammer throw, and pole vault when compared to women with the lowest testosterone. The study concluded that female athletes with high natural testosterone concentrations may have a competitive advantage over those with low testosterone competing in these specific track-and-field events. Thus, the quantitative relation between elevated testosterone and improved athletic performance should be considered when regulatory and governing bodies discuss the eligibility of women with hyperandrogenism in competitive events.

Thyroid Gland

The thyroid gland is located along the midline of the neck, immediately below the larynx. It secretes two important nonsteroid hormones, **triiodothyronine (T₃)** and **thyroxine (T₄)**, which regulate metabolism in general, and an additional hormone, calcitonin, which assists in regulating calcium metabolism.

The two metabolic thyroid hormones share similar functions. Triiodothyronine and thyroxine increase the metabolic rate of almost all tissues and can increase the body's basal metabolic rate by as much as 100%. These hormones also

- increase protein synthesis (including enzymes),
- increase the size and number of mitochondria in most cells,
- promote rapid cellular uptake of glucose,
- enhance glycolysis and gluconeogenesis, and
- enhance lipid mobilization, increasing FFA availability for oxidation.

Acute exercise causes the release of **thyrotropin (TSH)**, or thyroid-stimulating hormone) from the anterior pituitary. Thyroid-stimulating hormone controls the release of triiodothyronine and thyroxine, so the exercise-induced increase in TSH would be expected to stimulate the thyroid gland. Exercise increases plasma thyroxine concentrations, but a delay occurs between the increase in TSH concentrations during exercise and the increase in plasma thyroxine concentration. Furthermore, during prolonged submaximal exercise, thyroxine

concentration increases sharply, then remains relatively constant while triiodothyronine concentrations tend to decrease over time.

Adrenal Glands

The adrenal glands are situated directly atop each kidney and are composed of the inner adrenal medulla and the outer adrenal cortex. The hormones secreted by these two areas are quite distinct. The adrenal medulla produces and releases two hormones, **epinephrine** and norepinephrine, which are collectively referred to as **catecholamines**. Because of its origin in the adrenal gland, a synonym for epinephrine is **adrenaline**. When the adrenal medulla is stimulated by the sympathetic nervous system, approximately 80% of its secretion is epinephrine and 20% is norepinephrine, although these percentages vary with different physiological conditions. Circulating catecholamines have powerful effects similar to those of the sympathetic nervous system. Recall that these same catecholamines function as neurotransmitters in the sympathetic nervous system; however, the hormones' effects last longer because these substances are removed from the blood relatively slowly compared to the quick reuptake and degradation of the neurotransmitters. These two hormones prepare a person for immediate action, often called the fight-or-flight response.

Although some of the specific actions of these two hormones differ, the two work together. Their combined effects include

- increased heart rate and force of contraction,
- increased metabolic rate,
- increased glycogenolysis (breakdown of glycogen to glucose) in the liver and muscle,
- increased release of glucose and FFAs into the blood,
- redistribution of blood to the skeletal muscles,
- increased blood pressure, and
- increased respiration.

Release of epinephrine and norepinephrine is affected by a wide variety of factors, including psychological stress and exercise. Plasma concentrations of these hormones increase as individuals increase their exercise intensity. Plasma norepinephrine

concentrations increase markedly at intensities above 50% of $\dot{V}O_{2\max}$, but epinephrine concentrations do not increase significantly until the exercise intensity exceeds 60% to 70% of $\dot{V}O_{2\max}$. During long-duration steady-state exercise at a moderate intensity, blood concentrations of both hormones increase. When the exercise bout ends, epinephrine returns to resting concentrations within only a few minutes of recovery, but norepinephrine can remain elevated for several hours.

The adrenal cortex secretes more than 30 different steroid hormones, referred to as corticosteroids. These generally are classified into three major types: mineralocorticoids (discussed later in the chapter), glucocorticoids, and gonadocorticoids (sex hormones).

The **glucocorticoids** are essential to the ability to adapt to exercise and other forms of stress. They also help maintain fairly consistent plasma glucose concentrations even during long periods without ingestion of food. **Cortisol**, also known as hydrocortisone, is the major corticosteroid. It is responsible for about 95% of all glucocorticoid activity in the body. Cortisol

- stimulates gluconeogenesis to ensure an adequate fuel supply;
- increases mobilization of FFAs, making them more available as an energy source;
- decreases glucose utilization, sparing it for the brain;
- stimulates protein catabolism to release amino acids for use in repair, enzyme synthesis, and energy production;
- acts as an anti-inflammatory agent;
- depresses immune reactions; and
- increases the vasoconstriction caused by epinephrine.

We discuss cortisol's important role in exercise later in this chapter when we consider the regulation of glucose and fat metabolism.

Pancreas

The pancreas is located behind and slightly below the stomach. Its two major hormones are insulin and glucagon. The balance of these two opposing hormones provides the major control of plasma glucose

concentration. When plasma glucose is elevated (**hyperglycemia**), as occurs after a meal, the pancreas releases **insulin** into the blood. Among its actions, insulin

- facilitates glucose transport into the cells, especially muscle fibers;
- promotes glycogenesis; and
- inhibits gluconeogenesis.

Insulin's main function is to reduce the amount of glucose circulating in the blood. But it is also involved in protein and fat metabolism, promoting cellular uptake of amino acids and enhancing synthesis of protein and fat.

The pancreas secretes **glucagon** when the plasma glucose concentration falls below normal concentrations (**hypoglycemia**). The effects of glucagon generally oppose those of insulin. Glucagon promotes increased breakdown of liver glycogen to glucose (glycogenolysis) and increased gluconeogenesis, both of which increase plasma glucose levels.

During exercise lasting 30 min or longer, the body attempts to maintain plasma glucose concentrations; however, insulin concentrations tend to decline. The ability of insulin to bind to its receptors on muscle cells increases during exercise, due in large part to increased blood flow to muscle. This increases the body's sensitivity to insulin and reduces the need to maintain high plasma insulin concentrations for transporting glucose into the muscle cells. Plasma glucagon, on the other hand, shows a gradual increase throughout exercise. Glucagon primarily maintains plasma glucose concentrations by stimulating liver glycogenolysis. This increases glucose availability to the cells, maintaining adequate plasma glucose concentrations to meet increased metabolic demands. The responses of these hormones are usually blunted in trained individuals, and those who are well trained are better able to maintain plasma glucose concentrations.

Regulation of Carbohydrate Metabolism During Exercise

As we learned in [chapter 2](#), the heightened energy demands of exercise require that more glucose be made available to the muscles. Because glucose is stored in the body as glycogen, primarily in the

muscles and the liver, glycogenolysis must increase to free the glucose from this storage form. Glucose freed from the liver enters the blood to circulate throughout the body, allowing it access to active tissues. Plasma glucose concentration also can be increased through gluconeogenesis, the production of new glucose from noncarbohydrate sources like lactate, amino acids, and glycerol.

Regulation of Plasma Glucose Concentration

The plasma glucose concentration during exercise depends on a balance between glucose uptake by exercising muscles and its release by the liver. Four hormones work to increase the circulating plasma glucose:

- Glucagon
- Epinephrine
- Norepinephrine
- Cortisol

At rest, glucose release from the liver is facilitated by glucagon, which promotes both liver glycogen breakdown and glucose formation from amino acids. During exercise, glucagon secretion increases, as does the rate of catecholamine release from the adrenal medulla; these three hormones (glucagon, epinephrine, and norepinephrine) work in concert to further increase glycogenolysis. After a slight initial drop, cortisol concentration increases during the first 30 to 45 min of exercise. Cortisol increases protein catabolism, freeing amino acids to be used within the liver for gluconeogenesis. Thus, all four of these hormones can increase plasma glucose by enhancing the processes of glycogenolysis (breakdown of glycogen) and gluconeogenesis (making glucose from other substrates). In addition to the effects of the four major glucose-controlling hormones, GH increases mobilization of FFAs and decreases cellular uptake of glucose, so less glucose is used by the cells and more remains in circulation. The thyroid hormones promote glucose catabolism and fat metabolism.

The amount of glucose released by the liver depends on both exercise intensity and duration. As intensity increases, so does the rate of catecholamine release. This can cause the liver to release

more glucose than is being taken up by the active muscles. Consequently, during or shortly after an explosive, short-term sprint, blood glucose concentrations may be 40% to 50% above the resting value, since glucose is released by the liver at a greater rate than the rate of uptake by the muscles.

The greater the exercise intensity, the greater the catecholamine release, and thus the rate of glycogenolysis is significantly increased. This process occurs not only in the liver but also in the muscle. Glucose released from the liver enters the blood, where it becomes available to the muscle fibers. But the muscle has a more readily available source of glucose: its own glycogen stores. The muscle uses its own glycogen stores before using the plasma glucose during explosive, short-term exercise. Glucose released from the liver is not used as readily, so it remains in the circulation, elevating the plasma glucose. Following exercise, plasma glucose concentration decreases as the glucose enters the muscle to replenish the depleted muscle glycogen stores (glycogenolysis).

During exercise bouts that last for several hours, however, the rate of liver glucose release more closely matches the muscles' needs, keeping plasma glucose at or only slightly above the resting concentrations. As muscle uptake of glucose increases, the liver's rate of glucose release also increases. In most cases, plasma glucose does not begin to decline until late in the activity as liver glycogen stores become depleted, at which time the glucagon concentration increases significantly. Glucagon and cortisol together enhance gluconeogenesis, providing more fuel.

Figure 4.4 illustrates the changes in plasma concentrations of epinephrine, norepinephrine, glucagon, cortisol, and glucose during 3 h of cycling. Although the hormonal regulation of glucose remains intact throughout such long-term activities, the liver's glycogen supply may become limiting and the liver's rate of glucose release may be unable to keep pace with the muscles' rate of glucose uptake. Under this condition, plasma glucose may decline despite strong hormonal stimulation. Glucose ingestion during the activity can play a major role in maintaining plasma glucose concentrations.

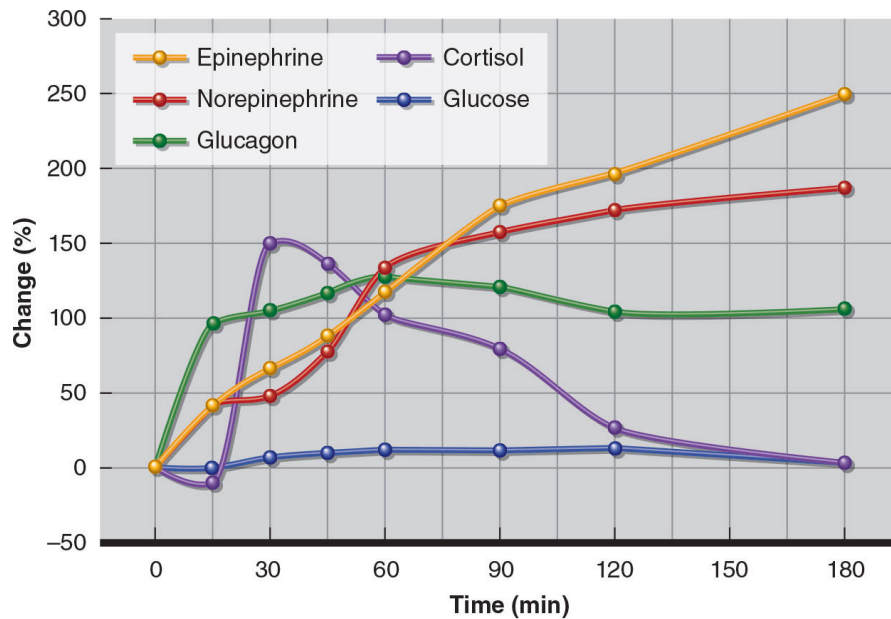


FIGURE 4.4 Changes (as a percentage of preexercise values) in plasma concentrations of epinephrine, norepinephrine, glucagon, cortisol, and glucose during 3 h of cycling at 65% $\dot{V}O_2$.



Glucose Uptake by Muscle

Merely releasing sufficient amounts of glucose into the blood does not ensure that the muscle cells will have enough glucose to meet their energy demands. Not only must the glucose be released and delivered to these cells, it also must be taken up by the cells. Transport of glucose through the cell membranes and into muscle cells is controlled by insulin. Once glucose is delivered to the muscle, insulin facilitates its transport into the fibers.

Surprisingly, as seen in [figure 4.5](#), plasma insulin concentration tends to decrease during prolonged exercise, despite a slight increase in plasma glucose concentration and glucose uptake by muscle. This apparent contradiction between the plasma insulin concentrations and the muscles' need for glucose serves as a reminder that a hormone's activity is determined not only by its concentration in the blood but also by a cell's sensitivity to that

hormone. In this case, the cell's sensitivity to insulin is at least as important as the concentration of circulating hormone. Exercise may enhance insulin's binding to receptors on the muscle fiber, thereby reducing the need for high concentrations of plasma insulin to transport glucose across the muscle cell membrane into the cell. This is important, because during exercise, four hormones are working to release glucose from its storage sites and create new glucose. High insulin concentrations would oppose their action, preventing this needed increase in plasma glucose supply.

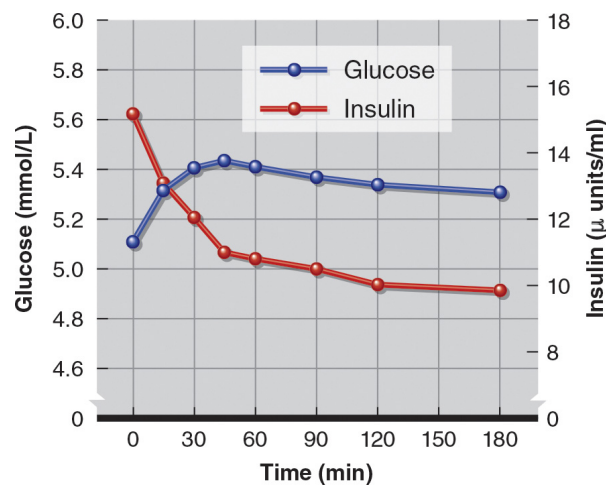


FIGURE 4.5 Changes in plasma concentrations of glucose and insulin during prolonged cycling at 65% to 70% of $\dot{V}O_2$. Note the gradual decline in insulin throughout the exercise, suggesting an increased sensitivity to insulin during prolonged effort.



CNS–Endocrine System Interaction

The central nervous system (CNS) integrates the activities of the nervous and endocrine systems. Therefore, it is not surprising that the CNS is involved in the regulation of carbohydrate metabolism through the sensing of hormones (especially insulin) and nutrients (including glucose, fatty acids, and amino acids).

The actions of insulin on the CNS were clarified through studies using a mouse model of insulin resistance, a condition commonly

associated with obesity.⁴ In this model, insulin signaling directly to neurons in the brain regulated how the tissues elsewhere around the body regulated glucose metabolism. Other studies have similarly demonstrated the important regulatory actions of the CNS on insulin's control of carbohydrate metabolism throughout the body. In these studies, the researchers directly injected glucose into areas of the brain to specifically stimulate receptors sensitive to glucose. Then they measured the hormones that regulate glucose metabolism as well as how much glucose was taken up and stored by the liver and muscle, respectively.⁷ By injecting glucose into the brain to examine central signaling and measuring glucose uptake throughout the body, they were able to show that the brain itself is indeed sensitive to glucose and helps to control the hormones released throughout the body in the regulation of carbohydrate metabolism.

Other hormones have a similar integration with the CNS. *Leptin* is a hormone that is released by adipose tissue in response to feeding, suppressing food intake. It also acts through specific CNS neurons called pro-opiomelanocortin (POMC) neurons to decrease glucose production in the liver since more glucose is not required after feeding. *Glucagon-like peptide 1* (GLP-1), a hormone released in the gut that signals β -cells in the pancreas to release insulin, also works through CNS POMC cells to decrease liver glucose production through both a decrease in gluconeogenesis and increased glycogenolysis. The integration of these hormonal effects through the CNS and subsequent peripheral actions is illustrated in [figure 4.6](#).

Within the brain itself, glucose regulation is particularly important because glucose is the only substrate that can be used for the brain's metabolism. Neuronal activity is tightly coupled with glucose utilization, and neurons preferentially use glucose derived from lactate (see [chapter 2](#)) as an oxidative fuel source.¹³ As in exercising muscle, lactate can be shuttled between cells in the brain to support oxidative metabolism.⁸ Together these findings illustrate the important role of the CNS in regulating hormones associated with carbohydrate metabolism and glucose homeostasis, both within the CNS and throughout the body.

Regulation of Fat Metabolism During Exercise

Free fatty acids are a primary source of energy at rest and during prolonged endurance exercise. They are derived from triglycerides through the action of the enzyme lipase, which breaks down triglycerides into FFA and glycerol. Although fat generally contributes less than carbohydrate does to muscles' energy needs during most bouts of exercise, mobilization and oxidation of FFAs are critical to performance in endurance exercise. During such prolonged activity, carbohydrate reserves become depleted, and muscle must rely more heavily on the oxidation of fat for energy production. When carbohydrate reserves are low (low plasma glucose and low muscle glycogen), the endocrine system can accelerate the oxidation of fats (lipolysis), thus ensuring that muscles' energy needs can be met.



Free fatty acids are stored as triglycerides in adipose tissue and within muscle fibers. Adipose tissue triglycerides, once broken down to release the FFAs, must be transported to the muscle fibers. The rate of FFA uptake by active muscle correlates highly with the plasma FFA concentration. Increasing this concentration would increase cellular uptake of the FFA. Therefore, the rate of triglyceride breakdown may determine, in part, the rate at which muscles use fat as a fuel source during exercise.

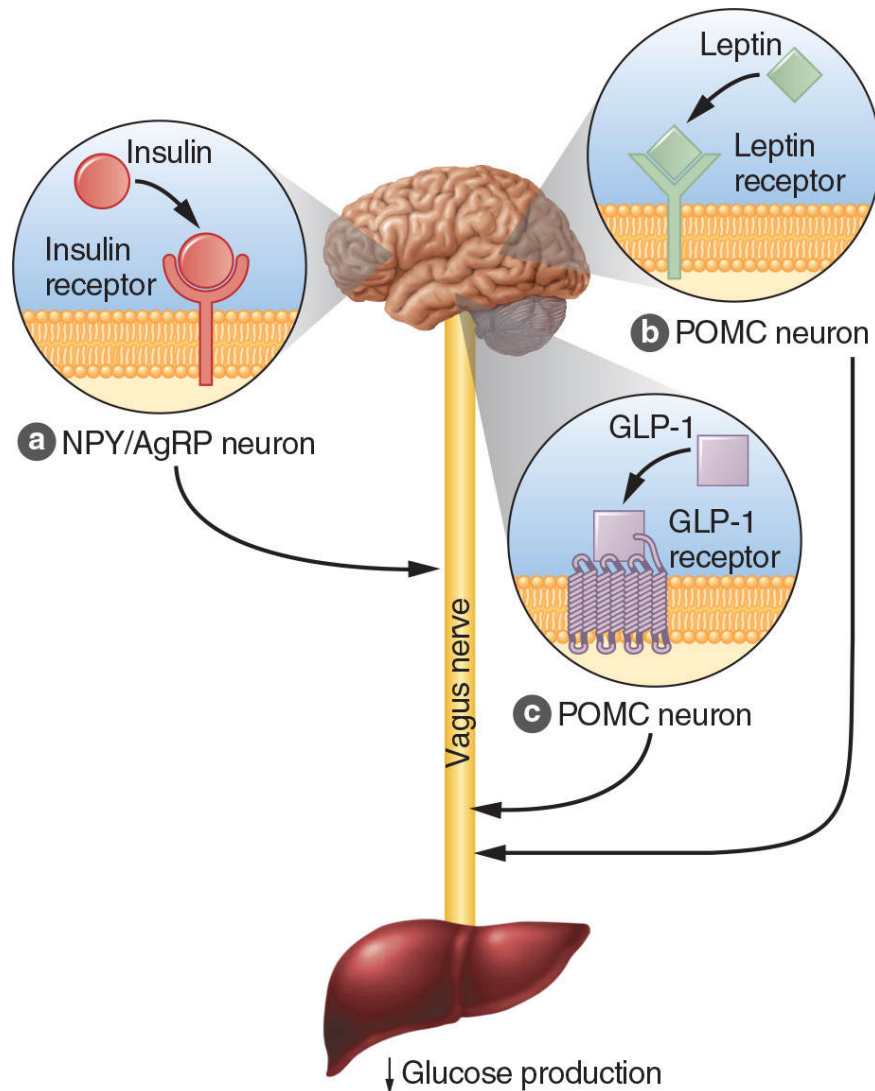


FIGURE 4.6 Hormones secreted throughout the peripheral tissues in the body, including the gastrointestinal tract and the pancreas, stimulate specific receptors in the hypothalamus to control glucose metabolism in the liver. (a) Insulin released by the β -cells in the pancreas acts through appetite-stimulating (NPY/AgRP) neurons in the arcuate nucleus of the hypothalamus. These neurons are stimulated by the peptide neurotransmitter neuropeptide Y (NPY) and release agouti-related peptide; insulin receptors are also present on these specialized neurons. (b and c) The pro-opiomelanocortin (POMC) neurons are stimulated by both leptin and glucagon-like peptide 1 (GLP-1). Together these hormones act on neurons in the brain, signaling through the vagus nerve to the liver to decrease glucose production.

Based on Lam et al. (2005).

The rate of lipolysis is controlled by at least five hormones:

- (Decreased) insulin
- Epinephrine
- Norepinephrine

- Cortisol
- Growth hormone

The major factor responsible for adipose tissue lipolysis during exercise is a fall in circulating insulin. Lipolysis is also enhanced through the elevation of epinephrine and norepinephrine. In addition to having a role in gluconeogenesis, cortisol accelerates the mobilization and use of FFAs for energy during exercise. Plasma cortisol concentration peaks after 30 to 45 min of exercise and then decreases to near-normal levels. But the plasma FFA concentration continues to increase throughout the activity, meaning that lipase continues to be activated by other hormones. The hormones that continue this process are the catecholamines and GH. The thyroid hormones also contribute to the mobilization and metabolism of FFAs but to a much lesser degree.

Thus, the endocrine system plays a critical role in regulating ATP production during exercise as well as controlling the balance between carbohydrate and fat metabolism.

In Review

- Plasma glucose concentration is increased by the combined actions of glucagon, epinephrine, norepinephrine, and cortisol. These hormones promote glycogenolysis and gluconeogenesis, thus increasing the amount of glucose available for use as a fuel source. This is important during exercise, particularly long-duration or high-intensity exercise, when blood glucose concentrations might otherwise decline.
- Insulin allows circulating glucose to enter the cells, where it can be used for energy production. But insulin concentrations decline during prolonged exercise, indicating that exercise increases cell sensitivity to insulin so that less of the hormone is required during exercise than at rest.
- When carbohydrate reserves are low, the body turns more to fat oxidation for energy and lipolysis increases. This process is facilitated by a decreased insulin concentration and increased concentrations of epinephrine, norepinephrine, cortisol, and GH.

Hormonal Regulation of Fluid and Electrolytes During Exercise

Fluid balance during exercise is critical for optimal metabolic, cardiovascular, and thermoregulatory function. At the onset of exercise, water shifts from the plasma volume to the interstitial and intracellular spaces. This water shift is specific to the amount of muscle that is active and the intensity of effort. Metabolic by-products begin to accumulate in and around the muscle fibers, increasing the osmotic pressure there. Water then moves passively into these areas by diffusion. Also, increased muscular activity increases blood pressure, which in turn drives water out of the blood (hydrostatic forces). In addition, sweating increases during exercise. The combined effect of these actions is that plasma volume decreases. For example, prolonged running at approximately 75% of $\dot{V}O_{2\max}$ decreases plasma volume by 5% to 10%. Reduced plasma volume can decrease blood pressure and increase the strain on the heart to pump blood to the working muscles. Both of these effects can impede athletic performance.

Endocrine Glands Involved in Fluid and Electrolyte Homeostasis

The endocrine system plays a major role in monitoring fluid levels and electrolyte balance, especially that of sodium. The two major endocrine glands involved in these processes are the posterior pituitary and the adrenal cortex. Additionally, the kidneys not only serve as the primary target organ for hormones released by these glands but also function as endocrine glands themselves.



Posterior Pituitary

The pituitary's posterior lobe is an outgrowth of neural tissue from the hypothalamus. For this reason, it is also referred to as the neurohypophysis. It secretes two hormones: **antidiuretic hormone (ADH)**—also called vasopressin or arginine vasopressin—and oxytocin. Both of these hormones are actually produced in the hypothalamus, travel through nerves, and are stored in vesicles within nerve endings in the posterior pituitary. These hormones are released into capillaries as needed in response to neural impulses from the hypothalamus.

Of the two posterior pituitary hormones, only ADH is known to play an important role during exercise. Antidiuretic hormone promotes

water conservation by increasing water reabsorption by the kidneys. As a result, less water is excreted in the urine, creating an “antidiuresis.”

Muscular activity and sweating cause electrolytes to become concentrated in the blood plasma as more fluid, compared to electrolytes, leaves the plasma. This is called **hemoconcentration**, and it increases the plasma **osmolality**. Osmolality refers to the ionic concentration of dissolved substances in the plasma. The presence of dissolved molecules and minerals in various body fluid compartments (i.e., intracellular, plasma, and interstitial spaces) generates an osmotic pressure or attraction to retain water within a compartment. The amount of osmotic pressure exerted by a body fluid is proportional to the number of molecular particles (osmoles, or Osm) in solution. A solution that has 1 Osm of solute dissolved in each kilogram (the weight of a liter) of water is said to have an osmolality of 1 osmole per kilogram (1 Osm/kg), whereas a solution that has 0.001 Osm/kg has an osmolality of 1 milliosmole per kilogram (1 mOsm/kg). Normally, body fluids have an osmolality of 300 mOsm/kg. Increasing the osmolality of the solutions in one body compartment generally causes water to be drawn away from adjacent compartments that have a lower osmolality (i.e., more water).

An increased plasma osmolality is the primary physiological stimulus for ADH release. The increased osmolality is sensed by osmoreceptors in the hypothalamus. A second and related stimulus for ADH release is a low plasma volume sensed by baroreceptors in the cardiovascular system. In response to either stimulus, the hypothalamus sends neural impulses to the posterior pituitary, stimulating ADH release. The ADH enters the blood, travels to the kidneys, and promotes water retention in an effort to dilute the plasma electrolyte concentration back to normal levels. This hormone’s role in conserving body water minimizes the extent of water loss and therefore the risk of severe dehydration during periods of heavy sweating and hard exercise. [Figure 4.7](#) illustrates this process.

Adrenal Cortex

A group of hormones called **mineralocorticoids**, secreted from the adrenal cortex, maintain electrolyte balance, especially that of sodium (Na^+) and potassium (K^+), in the extracellular fluids. **Aldosterone** is

the major mineralocorticoid, responsible for at least 95% of all mineralocorticoid activity. It works primarily by promoting renal reabsorption of sodium, thus causing the body to retain sodium. When sodium is retained, so is water, which follows the osmotic gradient; thus, aldosterone, like ADH, results in water retention. Sodium retention also enhances potassium excretion, so aldosterone plays a role in potassium balance as well. For these reasons, aldosterone secretion is stimulated by many factors, including decreased plasma sodium, decreased blood volume, decreased blood pressure, and increased plasma potassium concentration.



FIGURE 4.7 The mechanism by which antidiuretic hormone (ADH) helps to conserve body water.



The Kidneys as Endocrine Organs

Although the kidneys are not typically considered major endocrine organs, they do release two important hormones. The kidneys play a role in determining the aldosterone concentration in the blood. While the primary regulator of aldosterone release is plasma electrolyte concentration, a second set of hormones also determines aldosterone concentration and thus helps regulate body fluid balance. In response to a fall in blood pressure or plasma volume, blood flow to the kidneys decreases. Stimulated by activation of the sympathetic nervous system, the kidneys release **renin**. Renin is an enzyme that is released into the circulation, where it converts a molecule called angiotensinogen to angiotensin I. Angiotensin I is subsequently converted to its active form, angiotensin II, in the lungs with the aid of an enzyme, **angiotensin-converting enzyme (ACE)**. Angiotensin II stimulates aldosterone release from the adrenal cortex for sodium and water resorption at the kidneys. [Figure 4.8](#) shows the mechanism involved in renal control of blood pressure, the **renin-angiotensin-aldosterone mechanism**. In addition to stimulating aldosterone release from the adrenal cortex, angiotensin II causes blood vessels to constrict. Because ACE catalyzes the conversion of angiotensin I to angiotensin II, ACE inhibitors are sometimes prescribed for individuals with hypertension, since relaxation of the blood vessels lowers blood pressure.

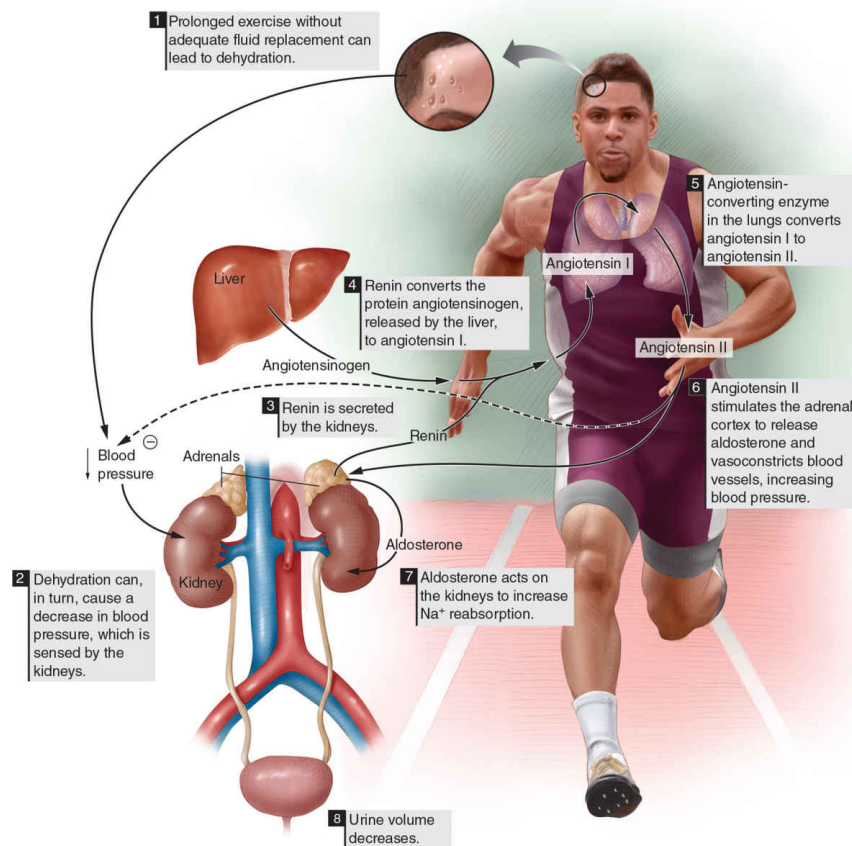


FIGURE 4.8 Water loss from plasma during exercise leads to a sequence of events that promotes sodium (Na^+) and water reabsorption from the renal tubules, thereby reducing urine production. In the hours after exercise when fluids are consumed, the elevated aldosterone concentration causes an increase in the extracellular volume and an expansion of plasma volume.



Recall that aldosterone's primary action is to promote sodium reabsorption in the kidneys. Because water follows sodium, this renal conservation of sodium causes the kidneys to also retain water. The net effect is to conserve the body's fluid content, thereby minimizing the loss of plasma volume while keeping blood pressure near normal.

Figure 4.9 illustrates the changes in plasma volume and aldosterone concentrations during 2 h of exercise. The hormonal influences of ADH and aldosterone persist for up to 48 h after exercise, reducing urine production and protecting the body from further dehydration.

The kidneys also release a hormone called erythropoietin. **Erythropoietin (EPO)** regulates red blood cell (erythrocyte) production by stimulating bone marrow cells. The red blood cells are essential for transporting oxygen to the tissues and removing carbon dioxide, so this hormone is extremely important in our adaptation to training and altitude.

Most athletes involved in heavy training have an expanded plasma volume, which dilutes various blood constituents. As proteins leave working muscle, they reenter the plasma through the lymphatic system, and water follows. This is a relatively short-term phenomenon, and new protein synthesis eventually supports this expanded plasma volume. During the early phases of plasma volume expansion, however, hemoglobin concentration decreases; that is, a **hemodilution** occurs.

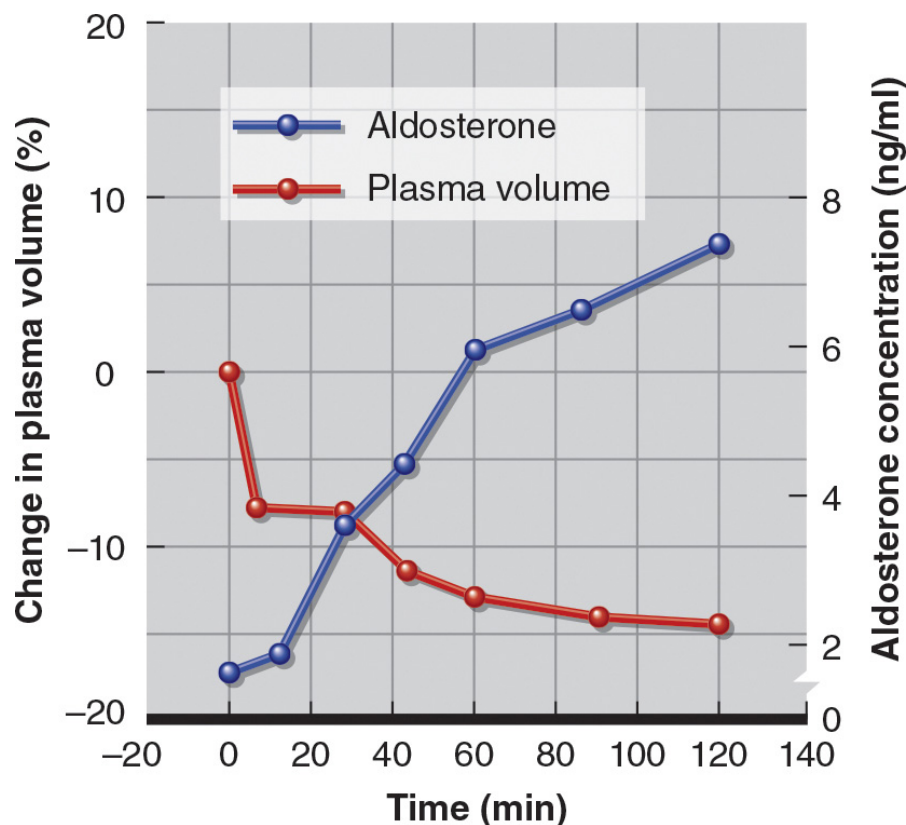
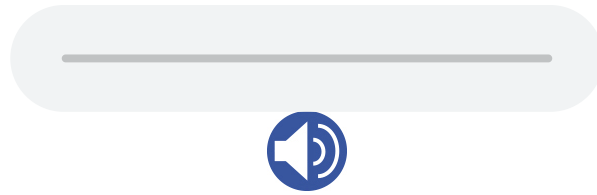


FIGURE 4.9 Changes in plasma volume and aldosterone concentration during 2 h of cycling exercise. Note that plasma volume declines rapidly during the first few minutes of exercise and then shows a smaller rate of decline despite large sweat losses. Plasma aldosterone concentration, on the other hand, increases rather steadily throughout the exercise.



The actual amount of hemoglobin has not changed; it is simply diluted. For this reason, some athletes who actually have normal hemoglobin concentrations may appear to be anemic as a consequence of Na^+ -induced hemodilution. This condition, not to be confused with true anemia, can be remedied with a few days of rest, allowing time for aldosterone concentrations to return to normal and for the kidneys to unload the extra Na^+ and water.

In Review

- Loss of fluid (plasma) from the blood results in a concentration of the constituents of the blood, a phenomenon referred to as hemoconcentration. Conversely, a gain of fluid in the blood results in a dilution of the constituents of the blood, which is referred to as hemodilution.
- The presence of dissolved particles in body fluid compartments generates an osmotic pressure or attraction to retain water. The osmotic pressure is proportional to the number of molecular particles in solution. A solution that has 1 osmole of solute dissolved in each kilogram (the weight of a liter) of water is said to have an osmolality of 1 osmole per kilogram (1 Osm/kg).
- Body fluids normally have an osmolality of 300 mOsm/kg. Increasing the osmolality of the solutions in one body compartment generally causes water to be drawn away from adjacent compartments.
- The two primary hormones involved in the regulation of fluid balance are ADH and aldosterone.
- Antidiuretic hormone is released in response to increased plasma osmolality. When osmoreceptors in the hypothalamus sense this increase, the hypothalamus triggers ADH release from the posterior pituitary. Low blood volume is a secondary stimulus for ADH release.

- Antidiuretic hormone acts on the kidneys, directly promoting water reabsorption and thus fluid conservation. As more fluid is resorbed, plasma volume increases and plasma osmolality decreases.
 - When plasma volume or blood pressure decreases, the kidneys release an enzyme called renin that converts angiotensinogen into angiotensin I, which later becomes angiotensin II in the lung circulation. Angiotensin II is a powerful constrictor of blood vessels and increases peripheral resistance, increasing the blood pressure.
 - Angiotensin II also triggers the release of aldosterone from the adrenal cortex. Aldosterone promotes sodium reabsorption in the kidneys, which in turn causes water retention, thus minimizing the loss of plasma volume.
-

Hormonal Regulation of Caloric Intake

The regulation of appetite, the sensations of hunger and satiety, and the feeling of fullness are part of a complex system that involves hormonal signaling from all over the body, including the gastrointestinal system and fat cells. Food intake is primarily under the control of the hypothalamus with some input from higher brain centers. The satiety area of the brain is located in the ventromedial nucleus, while the hunger center is located in the lateral hypothalamus. The hypothalamus, as it does for many aspects of homeostasis, integrates neural and hormonal signals for both the short- and long-term regulation of eating behavior and calorie intake.

Hormones that influence these brain centers are synthesized in, and released from, peripheral tissues including the gut and fat cells (adipocytes). These hormones can be categorically split into those that are anorexigenic, meaning that they suppress appetite, and those that are orexigenic, meaning that they stimulate appetite. The main hormones that regulate appetite and satiety are cholecystokinin, leptin, peptide YY, GLP-1, insulin, and ghrelin.

Gastrointestinal Tract Hormones

Short-term control of food intake is regulated by plasma concentrations of nutrients including amino acids, glucose, and lipids. However, another significant influence on short-term regulation of food intake involves hormones released in the gastrointestinal (GI) tract. Gastrointestinal distention caused by a full stomach triggers the

release of the hormone **cholecystokinin (CCK)**, which stimulates afferent fibers of the vagus nerve to send signals to the brain to suppress hunger. In addition, other hormones including GLP-1 and peptide YY (PYY) are secreted from the large and small intestines during and after eating. These hormones travel through the blood to the brain where they suppress hunger. Peptide YY also acts on the hypothalamus to inhibit gastric motility. Insulin released from the pancreas in response to eating also acts as a satiety hormone.

RESEARCH PERSPECTIVE 4.2

Endurance Training for More Red Blood Cells

The direct association between endurance training and increased red blood cell volume was first discovered in 1949.⁶ This adaptation to endurance training, called erythropoiesis, increases oxygen delivery to the exercising muscle by increasing the number of red blood cells available to carry oxygen in addition to increasing the blood volume pumped with every beat of the heart. Increasing red blood cell volume is a fundamental component of the increase in aerobic capacity (maximal oxygen uptake) that occurs with regular endurance exercise training (discussed further in [chapter 11](#)). Although erythropoiesis is a central mechanism by which endurance training adaptations occur, relatively little is known about how red blood cell volume expansion is regulated during repeated bouts of endurance training.

A recent study conducted in Switzerland examined erythropoiesis and its physiological regulators during an 8-week endurance-training program in a group of healthy young men and women.¹² Researchers measured body composition, heart rate, blood pressure, maximal exercise capacity, total blood volume, red blood cell volume, and erythropoietin (a hormone that stimulates red blood cell production) concentrations in the blood throughout the 8-week training period. At the end of the training period, average red blood cell volume had doubled and maximal exercise capacity had increased by ~10%. Across the 8 weeks of training, total blood volume increased during week 2 and remained high throughout; erythropoietin also increased in week 2 but subsequently returned to baseline values by week 4. Red blood cell volume was increased by week 4 and continued to increase through week 8, coming after the preceding increases in blood volume and circulating erythropoietin concentration. By week 8, exercise-induced increases in erythropoietin concentration had ended but red blood cell volume continued to increase, suggesting that there are still unexplained mechanisms that control erythropoiesis. These findings provided novel insight into the time course of expansion of red blood cell volume as an adaptation to endurance training, while at the same time uncovering new questions for future research.

Conversely, the hormone ghrelin is secreted from the stomach and pancreas when the stomach is empty; it can be thought of as a hunger hormone. **Ghrelin** is transmitted through the blood to the brain where it crosses the blood–brain barrier to act on the hunger areas in the lateral hypothalamus. After eating, ghrelin concentrations decrease.

Adipose Tissue as an Endocrine Organ

In addition to hormones secreted by the stomach and intestines to signal hunger or fullness, additional hormones are secreted by adipocytes (fat cells) that likewise act on the hunger and satiety centers in the hypothalamus. Because the level of these hormones depends on the amount of adipose tissue in the body, which changes slowly, these hormones are more involved in the long-term regulation of food intake. The hormone **leptin** is primarily secreted by fat cells and acts on receptors in the hypothalamus to decrease hunger. Leptin is also an indicator of energy balance, as its circulating concentrations are proportional to body fat. A simple schematic of how leptin and ghrelin interact to modify appetite and satiety is presented in [figure 4.10](#).

A great deal has been discovered about what leptin does in terms of energy balance from a mouse model using mice that lack the ability to make leptin in their fat cells. These mice have a voracious appetite and are massively obese. In obese humans, circulating concentrations of leptin are elevated, but many obese humans are leptin resistant. This suggests that despite an elevated signal that they are in an overfed state, the signal is not being transmitted through the hypothalamus to initiate the sensations of satiety. Interestingly, obese humans also appear to have a dampened ghrelin signal. Researchers are only beginning to understand how hormonal appetite signaling changes with weight gain and obesity. This is critical in order to determine how best to treat obesity, as well as how exercise may influence appetite and satiety hormones.

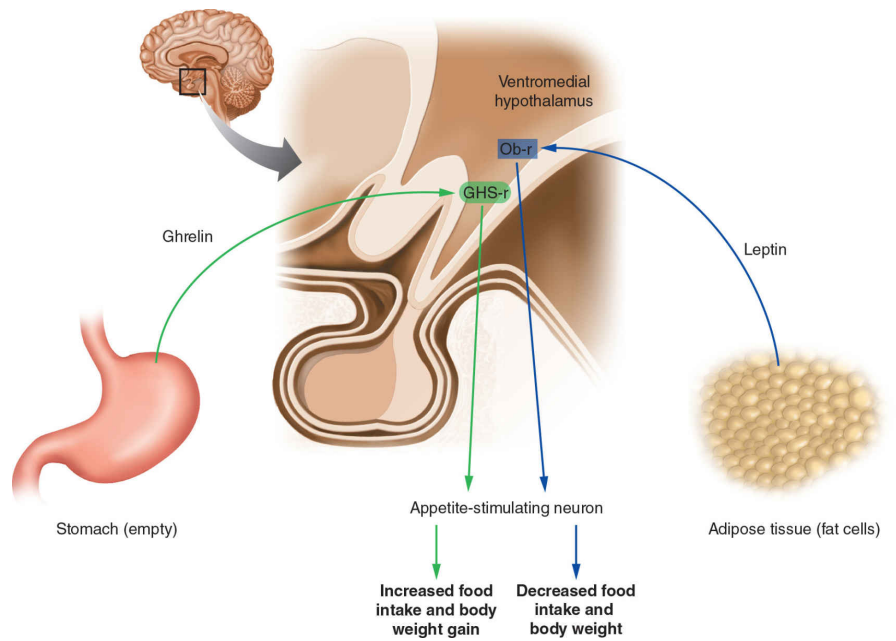


FIGURE 4.10 Hormonal regulation of appetite and satiety by ghrelin and leptin. Acting through specific hypothalamic receptors (GH secretagogue receptor, or GHS-r, for ghrelin and obesity receptor, or Ob-r, for leptin), ghrelin increases, and leptin decreases, appetite.



RESEARCH PERSPECTIVE 4.3

Does Environmental Temperature Alter the Hormones That Control Appetite?

The interactions among exercise, appetite, and energy intake are important for the control and maintenance of energy homeostasis and body weight. Scientifically, these interactions have received widespread attention because they may hold the key to treating excess weight gain and obesity. Leptin and ghrelin are hormones that regulate the perception of hunger and lead to changes in appetite. Leptin (the “satiety hormone”) decreases energy intake, while ghrelin (the “hunger hormone”) increases energy intake. Both exercise and exposure to extreme temperatures can affect the concentrations of these

appetite-regulating hormones. Circulating ghrelin concentration and perception of hunger both decrease immediately after a single bout of moderate- to high-intensity exercise but have no influence on the total energy intake throughout the day. Environmental temperature has an impact on resting metabolic rate. Indigenous populations who live in polar climates have elevated basal metabolic rates, while those who live in tropical climates have decreased basal metabolic rates. Additionally, exercise in a hot environment reduces appetite, while exercise in the cold stimulates appetite; however, it is unknown if these effects involve changes in circulating leptin or ghrelin.

Recently, a group of researchers at the University of Nebraska Omaha conducted an experiment to examine how exercise in different environmental temperatures would affect the leptin and ghrelin responses to exercise.⁹ Research subjects completed three separate 1 h bouts of cycling in hot (33 °C [91°F]), neutral (20 °C [68 °F]), and cold (7 °C [45 °F]) air temperatures. The research team measured leptin and ghrelin in blood samples collected preexercise, immediately postexercise, and after a 3 h recovery. Similar to previous studies, circulating leptin was increased immediately after exercise and remained elevated 3 h later. Circulating ghrelin concentrations did not change. Although the researchers hypothesized that there would be larger increases in leptin after exercise in the heat and larger increases in ghrelin after exercise in the cold, there was no effect of air temperature on any hormone measurements. The conclusion from this study was that environmental temperature does not alter the leptin or ghrelin responses to short bouts of aerobic exercise. Future research studies are needed to determine what other variables might affect regulatory hormone and hunger responses following exercise in extreme environments.

Effects of Acute and Chronic Exercise on Satiety Hormones

Acute bouts of moderate- to vigorous-intensity exercise temporarily suppress appetite, likely by decreasing ghrelin and increasing GLP-1 and PYY released from the GI tract.¹⁴ These hormonal changes are most pronounced with aerobic exercise and are not observed after resistance exercise training.³

With chronic exercise training comes a shift in energy balance due to the calorie deficit induced by exercise. This is accompanied by a partial compensation to increase hunger and therefore caloric intake through changes in the appetite-regulating hormones. Several studies have observed an increase in plasma PYY concentrations after exercise training, which would be consistent with improved satiety. Counterintuitively, the hunger hormone ghrelin does not change in people who do not lose weight during exercise training but increases

significantly in those who do lose weight.¹⁰ In general, appetite and satiety hormones are sensitive to the total energy balance that is modulated by regular exercise. It has been suggested that for elite athletes who need to monitor their energy balance, measures of circulating leptin and ghrelin may help to determine when the athlete is overtraining and may help predict states of energy deficit.⁵

IN CLOSING

In this chapter, we focused on the role of the endocrine system in regulating some of the many physiological processes that accompany exercise. We discussed the role of hormones in regulating the metabolism of glucose and fat for energy metabolism and the role of other hormones in maintaining fluid balance. We touched on some of the relatively new findings about how hormones regulate appetite and calorie consumption. We next look at the related topics of energy expenditure and fatigue during exercise.

KEY TERMS

adrenaline
aldosterone
angiotensin-converting enzyme (ACE)
antidiuretic hormone (ADH)
autocrines
catecholamines
cholecystokinin (CCK)
cortisol
cyclic adenosine monophosphate (cAMP)
direct gene activation
downregulation
epinephrine
erythropoietin (EPO)
ghrelin
glucagon
glucocorticoids
growth hormone (GH)
hemoconcentration
hemodilution
hormone
hyperglycemia
hypoglycemia
inhibiting factors
insulin
insulin resistance
insulin sensitivity
leptin
mineralocorticoids
nonsteroid hormones

osmolality
prostaglandins
releasing factors
renin
renin-angiotensin-aldosterone mechanism
second messenger
steroid hormones
target cells
thyrotropin (TSH)
thyroxine (T_4)
triiodothyronine (T_3)
upregulation

STUDY QUESTIONS

1. What is an endocrine gland, and what are the functions of hormones?
2. Explain the difference between steroid hormones and nonsteroid hormones in terms of their actions at target cells.
3. How can hormones have very specific functions when they reach nearly all parts of the body through the blood?
4. What determines plasma concentrations of specific hormones? What determines their effectiveness on target cells and tissues?
5. Define the terms *upregulation* and *downregulation*. How do target cells become more or less sensitive to hormones?
6. What are second messengers, and what role do they play in hormonal control of cell function?
7. Briefly outline the major endocrine glands, their hormones, and the specific action of these hormones.
8. Which of the hormones outlined in question 7 are of major significance during exercise?
9. What hormones are involved in the regulation of metabolism during exercise? How do they influence the availability of carbohydrates and fats for energy during exercise lasting for several hours?
10. Discuss how the central nervous system helps integrate glucose regulation and the hormones involved in this process.
11. Describe the hormonal regulation of fluid balance during exercise.
12. Discuss the sources and function of the hormones cholecystokinin, leptin, and ghrelin, and explain how they are interrelated.

STUDY GUIDE ACTIVITIES

In addition to the activities listed in the chapter opening outline, two other activities are available in the web study guide, located at

www.HumanKinetics.com/PhysiologyOfSportAndExercise

The **KEY TERMS** activity reviews important terms, and the end-of-chapter **QUIZ** tests your understanding of the material covered in the chapter.

